AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

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(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals
12. Individual Member Contribution

Problem Statement

The lack of integrated, real-time public transport information makes local bus travel unpredictable and discourages people from choosing buses over private vehicles for commuting within the district

Problem Understanding

- The problem primarily affects newcomers, tourists, college students, daily commuters, and occasional bus passengers who struggle to find reliable information about local bus routes, timings, fares, and boarding points. People unfamiliar with the area may find it difficult to identify the right bus or plan their journey efficiently.

- The problem is caused by the lack of integrated and reliable public transport information, including bus routes, schedules, fares, and arrival times. Limited access to real-time bus data, unpredictable arrival times, inconsistent stopping behaviour, and the absence of a single convenient platform make it difficult for passengers to plan their journeys, especially in unfamiliar areas.

- The lack of reliable bus information causes users to experience uncertainty, longer waiting times, missed buses, and difficulty planning their journeys, especially in unfamiliar areas. Students, commuters, and tourists may become frustrated and choose private transportation instead. Bus operators and public transport authorities may also face reduced passenger confidence and difficulty encouraging greater use of public transport.

Existing Gaps & Challenges

- Existing public transport apps may not provide all essential local bus information in one place, making it difficult for passengers to access route details, timetables, fares, and reliable arrival-time estimates, particularly when real-time data is unavailable.
- Many public transport solutions depend on the availability of real-time data, which may be limited or unavailable for certain local bus routes. This makes it difficult for passengers to obtain accurate arrival updates and reliable journey estimates.
- We cannot control bus arrival times, drivers' stopping behaviour, or the availability and accuracy of real-time transport data. It can only provide guidance based on the information available.

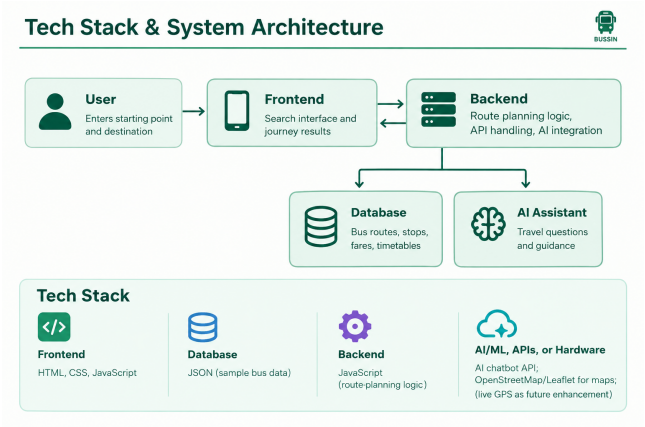
Our Proposed Solution

- This local bus journey planning app that helps users find suitable bus routes, check available timetables and fares, estimate journey durations, and identify boarding and drop-off points. It also features an AI travel assistant to guide users through unfamiliar routes. The app will use available, verified transport information and clearly indicate when real-time data is unavailable, making bus travel simpler and more convenient.

- Users enter their starting point and destination in the app which then uses available bus route, timetable, and fare information to suggest suitable routes, estimate journey duration, and provide boarding and drop-off guidance. An AI travel assistant helps answer travel-related questions, while real-time updates are shown wherever reliable data is available. If live data is unavailable, the app provides available schedule-based information instead.

- It makes local bus travel simpler, more convenient, and less confusing by bringing essential route, timetable, fare, and journey-planning information into one app. It helps users save time, make informed travel decisions, and navigate unfamiliar places with greater confidence.

Technical Approach



Users enter their source and destination, and the app uses available bus route, timetable, and fare data to suggest suitable journeys. It displays estimated travel times and boarding guidance, while an AI assistant helps answer travel-related questions. If real-time data is unavailable, the app relies on available schedule-based information.

Tech Stack

Frontend: HTML, CSS, JavaScript

Backend: JavaScript route-planning logic

Database: JSON sample bus data

AI/ML, APIs, or Hardware: AI chatbot API, OpenStreetMap/Leaflet for maps

Innovation & Uniqueness

- The app combines essential local bus information, route comparison, journey-planning guidance, and an AI travel assistant in one platform. Unlike solutions that rely heavily on live tracking, it remains useful through available route and timetable information even when real-time data is unavailable.
- The app addresses the lack of accessible and integrated bus information by bringing essential journey-planning features into one platform. It does not depend entirely on real-time tracking, making it more practical to implement with available transport data while helping users navigate unfamiliar routes with greater ease and confidence.
- The app combines route planning, fare information, journey estimates, and AI-powered travel assistance in one platform while remaining useful even when real-time bus data is unavailable.

Feasibility & Scalability

- **Feasibility:**The app can be developed using HTML, CSS, and JavaScript for the interface, along with verified sample bus-route, timetable, and fare data. Tools such as VS Code, Leaflet, and an AI API can be used to build the route planner, map, and travel assistant. The initial prototype can focus on a small number of local routes, with real-time tracking added later if reliable data becomes available.
- **Scalability:** The app can initially cover a small number of local bus routes and gradually expand to more routes, districts, and cities as reliable transport data becomes available. Future upgrades could include real-time bus tracking, personalized notifications, crowd-level updates, and support for more languages, allowing the app to serve a wider range of passengers.
- **Risks:** The main risks include limited access to reliable bus data, inaccurate timetable information, unpredictable bus stopping behaviour, and low user adoption. These can be mitigated by starting with selected routes, verifying available information, clearly labelling estimates, and gathering user feedback to improve the app. Real-time tracking can be introduced later if reliable data becomes available.

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases (compulsorily 3 use cases must be there)

- 1. A Student Travelling to a New College: A student unfamiliar with the area uses the app to find the correct bus route, check available timings and fares, and identify where to board and get off.
- 2. A Tourist Exploring a New City: A tourist enters a destination and compares available bus routes, estimated journey durations, and fares to choose a suitable way to travel.
- 3. A Daily Commuter Planning a Journey: A commuter checks timetable information before leaving home, compares alternative routes, and uses the app's travel assistant to clarify route or stop details.
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- The app helps newcomers, tourists, and daily commuters plan their bus journeys with greater ease and confidence. By providing bus routes, available timetables, fare information, estimated journey durations, and boarding guidance in one place, it reduces confusion and helps users make informed travel decisions. The app also helps users compare routes, save time when planning trips, and navigate unfamiliar areas more conveniently. Its AI travel assistant offers additional guidance, making local public transportation more accessible and user-friendly.

- The app helps passengers navigate local bus services more easily, improving accessibility and reducing travel-related confusion. By providing fare information and route comparisons, it helps users make cost-effective travel decisions. Encouraging greater use of public buses may also reduce dependence on private vehicles, helping to ease traffic congestion and lower emissions.

- It aims to help users find suitable bus routes, access available fare and timetable information, and plan journeys more efficiently.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 11	Target 11.2	Sustainable Cities and Communities	This app helps users access transport information and plan journeys more easily, supporting accessible and sustainable transportation.
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	The app uses digital technology to improve access to local bus information and support smarter transport services.

Individual Member Contribution

Member	Contribution or responsibility
R Rithikaa	Research and PPT
Sarvesh Subramanian S	Research and PPT
Anushka Acharjee	Research and PPT
Aachitya Singh	Research and PPT

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